EEE 474 - Foundations of Magnetic Resonance Imaging
B_0 Measurement of an Orange

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1 Introduction

The main B_0-field of MRI machines are rarely constant and homogeneous. Instead they feature an error component commonly in the order of ppm compared to the main field. If these inhomogeneities become too large they may distort the image. Hence it is important to measure and tune these errors to make them as little as possible. We can measure these inhomogeneities and tune the MRI scanner accordingly. This process is called shimming and it is crucial to acquiring accurate images.

These inhomogeneities may be due to imperfections in the generating coil or due to changes in the magnetic field arising from boundary conditions at tissue interfaces, where differences in magnetic susceptibility cause local variations in the B_0 field.

2 Theory

The effects of B_0 on the experiments' data are given as [1]:

$$s(t) = \int\int m(x,y)e^{-i2\pi f_e TE}e^{-i2\pi[k_x x + k_y y]} \tag{1}$$

We want to find f_e. To do this we can cancel out the $m(x,y)$ components. For two different experiments we can isolate the data as:

$$\mathcal{F}^{-1}_{(2)}\{s_1(t)\} = m(x,y)e^{-i2\pi f_e TE_1}, \quad \mathcal{F}^{-1}_{(2)}\{s_2(t)\} = m(x,y)e^{-i2\pi f_e TE_2} \tag{2}$$

We can obtain f_e as:

$$\angle\left(\mathcal{F}^{-1}_{(2)}\{s_1(t)\}\mathcal{F}^{-1}_{(2)}\{s_1(t)\}^*\right) = \angle\left(m(x,y)e^{-i2\pi f_e TE_1}m(x,y)^* e^{i2\pi f_e TE_2}\right) = 2\pi f_e(TE_1 - TE_2) \tag{3}$$

Multiplying with the conjugate and then taking the phase is equivalent to subtracting the phases of individual experiments as:

$$\angle\left(m(x,y)e^{-i2\pi f_e TE_1}\right) - \angle\left(m(x,y)e^{-i2\pi f_e TE_2}\right) \tag{4}$$

However, multiplying prevents phase discontinuities at $-\pi$ and π when processing the data in MATLAB. This can also be prevented with MATLAB's *unwrap()* function.

3 Methodology

We need to take the image with different TE values to calculate the error term. For this purpose Gradient Echo was used. Spin Echo was not preferred for this purpose as spin echo would also counteract the phase shifts caused by the inhomogeneities in B_0. If TE is chosen large, the image becomes T_2^* weighted.

Another pulse sequence called gradient field mapping was used. Which is automatically maps the MRI scanners B_0 field. Hence, it acted as a baseline to compare the results against.

4 Imaging & Images

The chosen fruit was an orange because of its size and water content. The fruit's cross section both imaged by the MRI scanner and photographed can be seen in Figure 1.

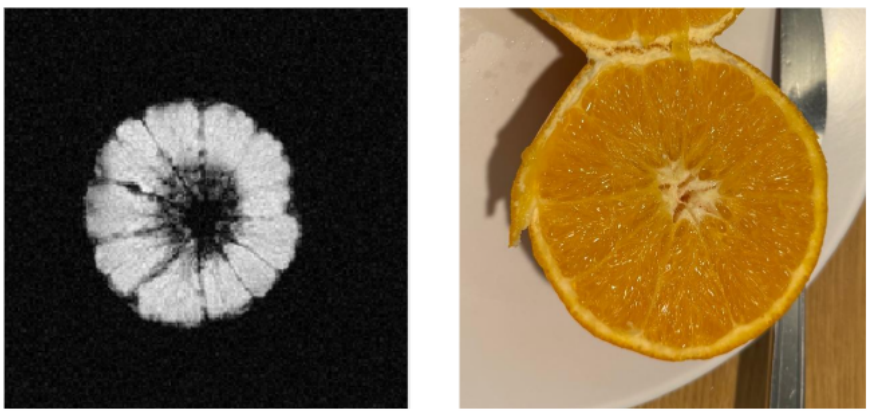


Figure 1: Comparison of MR Image and Photograph of the Orange

The imaging parameters were chosen as:

- TR = 500 ms
- TE = 5,10,15,20 ms
- FoV = 150 mm
- Slices = 5
- Average = 2
- Flip Angle = 45°
- Resolution = 192
- Bandwidth = 260 Hz/px
- RF Spoiling: ON

As mentioned, if the echo times (TE) is chosen too large, the images become T_2^* weighted. We can compare images with different TE values in Figure 2.

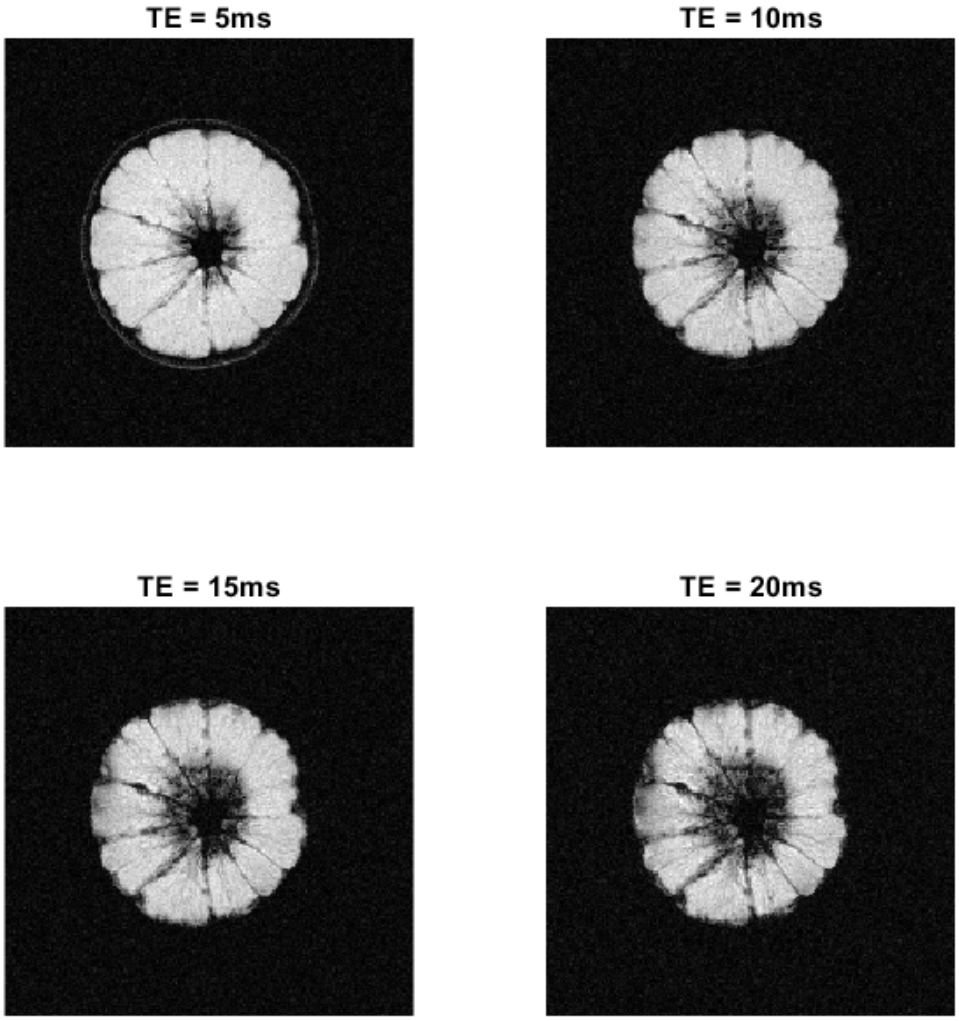


Figure 2: Images with Different TE Values

Images for other slices are available in Appendix A.

5 B_0 Mapping

To obtain the B_0 maps, we need to take the inverse Fourier Transforms of $s(t)'s$. We will then apply the formula discussed in §2. For $TE_1 = 5$ ms, $TE_2 = 10$ ms, the resulting B_0 map is given in Figure 3. The regions outside the orange consist of pure noise. This makes the results difficult to observe reliably. A mask can be used to flatten the voxels for which the signal intensity is below a certain threshold. Applying this mask to B_0 allows us to suppress noise-dominated regions and restrict the analysis to voxels with a sufficient signal-to-noise ratio, thus improving the reliability and interpretebility of the estimated field inhomogeneities. Masked version is given in Figure 4.

The sequence "Gradient Field Mapping" also provided the B_0 maps as a reference point to validate our results again. The results of this sequence can be found in Appendix B.

Note that these B_0 maps are given for the third slice.

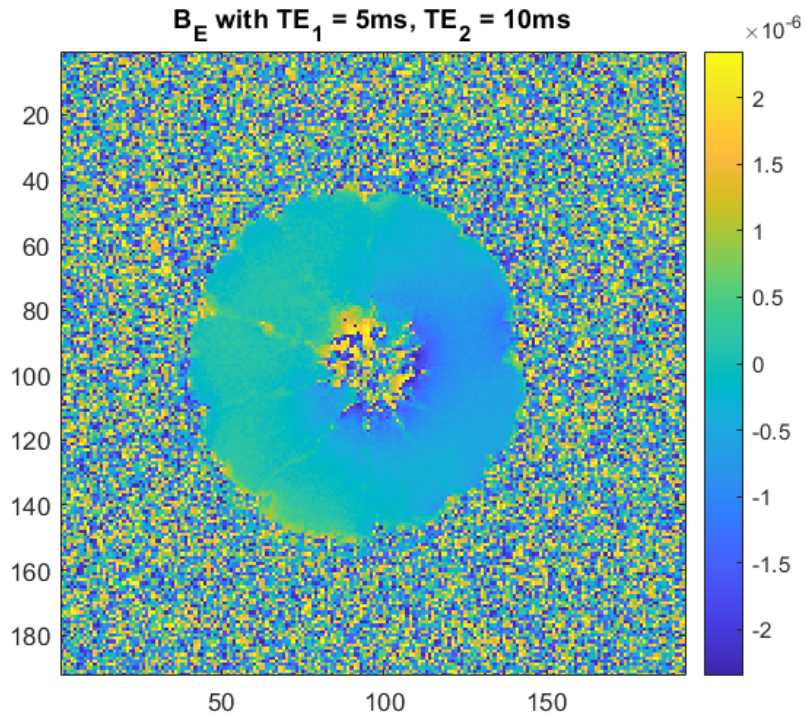


Figure 3: Unmasked B_0 Map for $TE_1 = 5$ ms, $TE_2 = 10$ ms

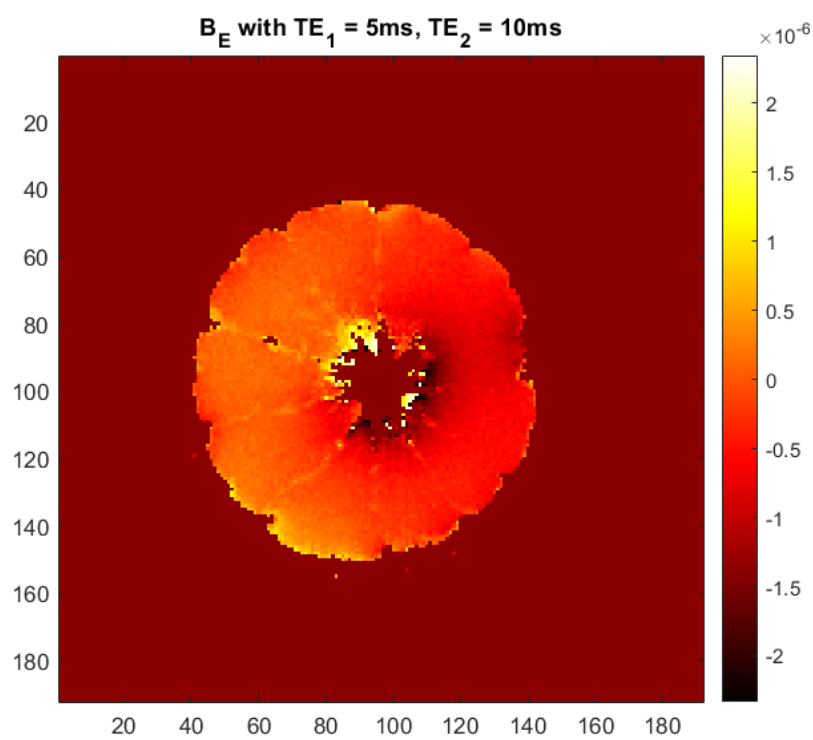


Figure 4: Masked B_0 Map for $TE_1 = 5$ ms, $TE_2 = 10$ ms

We can observe that the field inhomogeneities range from $-2.5\mu T, 2.5\mu T$.

6 Noise Analysis

The measurements outside the orange are just noise. The noise's phase is distributed uniformly along all angle values. Which can be shown as:

Let the complex baseband noise be

$$n = n_I + jn_Q$$

where n_I and n_Q are independent and identically distributed Gaussian random variables,

$$n_I, n_Q \sim \mathcal{N}(0, \sigma^2).$$

The joint probability density function (PDF) is therefore

$$f_{n_I,n_Q}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right).$$

Transforming to polar coordinates $x = r\cos\phi$, $y = r\sin\phi$, with Jacobian determinant r, yields

$$f_{r,\phi}(r, \phi) = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right).$$

The marginal PDF of the phase is obtained by integrating over r:

$$f_\phi(\phi) = \int_0^\infty f_{r,\phi}(r, \phi)\, dr = \frac{1}{2\pi}, \quad \phi \in (-\pi, \pi].$$

Thus, the phase of the complex Gaussian noise is uniformly distributed.

This makes the resulting error distributed uniformly as well! We can subtract the orange from the image and see the noise and its distribution in Figure 5.

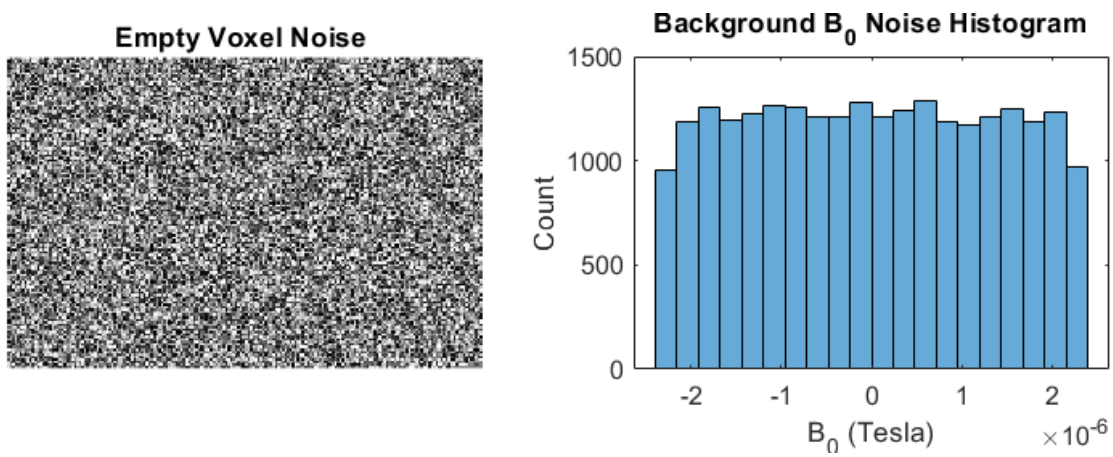


Figure 5: Noise Distribution

The variance of the noise can be calculated to be 1.8×10^{-12}. Which matches the theoretical expectation of 2.08×10^{-12} which was calculated via the formula for uniform distribution.

For voxels inside the orange, the phase noise is due to the quadrature component of the noise since the in-phase component does not alter the phase. Hence, the complex data no longer exhibits circular symmetry about the origin. As a result, the phase distribution is concentrated around the true phase ϕ_0, and noise manifests primarily as phase jitter. For high-SNR, this phase error is dominated by the quadrature noise component and remains small compared to the uniformly distributed phase observed in noise-only regions.

7 Conclusion

In this project, spatial variations of the main magnetic field B_0 were measured using a gradient echo sequences with different TE values, and phase–based field mapping. By acquiring images at different echo times and exploiting the phase shifting between them, the corresponding B_0 inhomogeneities were measured. Another sequence called gradient field mapping sequence was additionally used as a reference to validate the results.

In the unmasked B_0 maps, noise-dominated regions outside the orange was investigated and masked. Applying a mask significantly improved the interpretability of the field maps by suppressing voxels with insufficient signal intensity. After masking, the observed field variations within the orange were on the order of a few microtesla, which is consistent with the expected magnitude of B_0 inhomogeneities in MRI scanners.

A detailed noise analysis showed that, in regions without signal, the complex noise exhibits circular symmetry, resulting in a uniformly distributed phase. This leads to large apparent B_0 fluctuations that are purely noise-driven. In contrast, for voxels containing signal, the presence of a dominant deterministic component breaks this symmetry, and noise primarily manifests as small phase perturbations governed by the quadrature noise component. This explains the observed difference between noise-only and signal-present regions in the B_0 maps and highlights the importance of masking in phase-based field estimation.

Overall, the results demonstrate that dual-echo phase mapping provides a reliable method for estimating B_0 inhomogeneities when combined with appropriate noise suppression techniques. The findings also emphasize the fundamental role of phase noise characteristics in determining the accuracy and robustness of B_0 measurements.

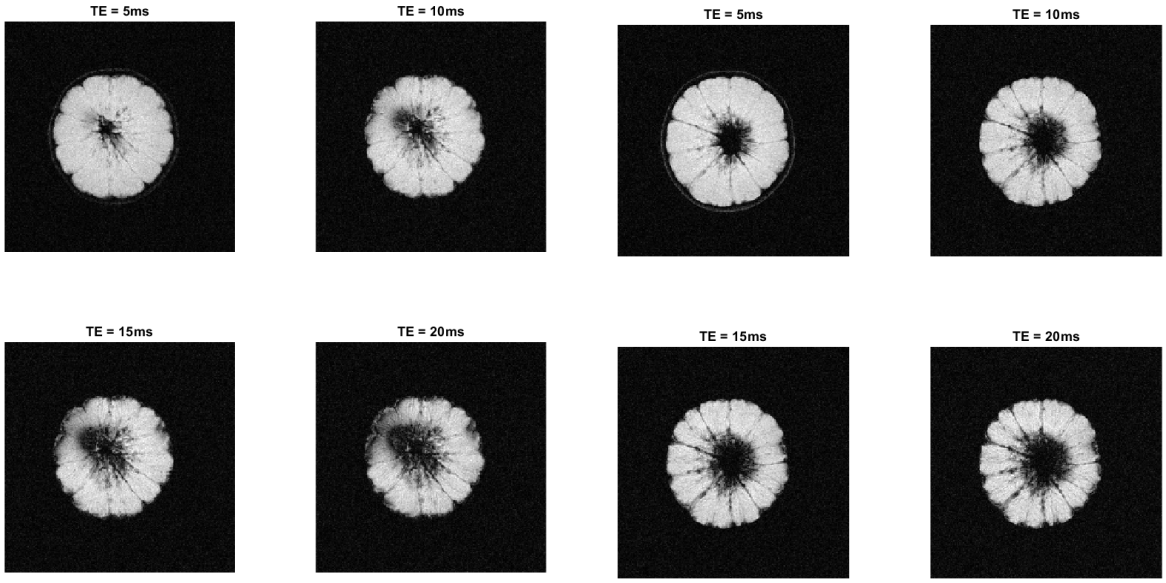
References

[1] D. G. Nishimura, *Principles of Magnetic Resonance Imaging.* Stanford University, 2010.

[2] P. Ehses, "mapvbvd: Matlab/octave tool to read twix raw data from siemens mri scanners," GitHub repository, 2025, accessed: Dec. 22, 2025. [Online]. Available: https://github.com/pehses/mapVBVD

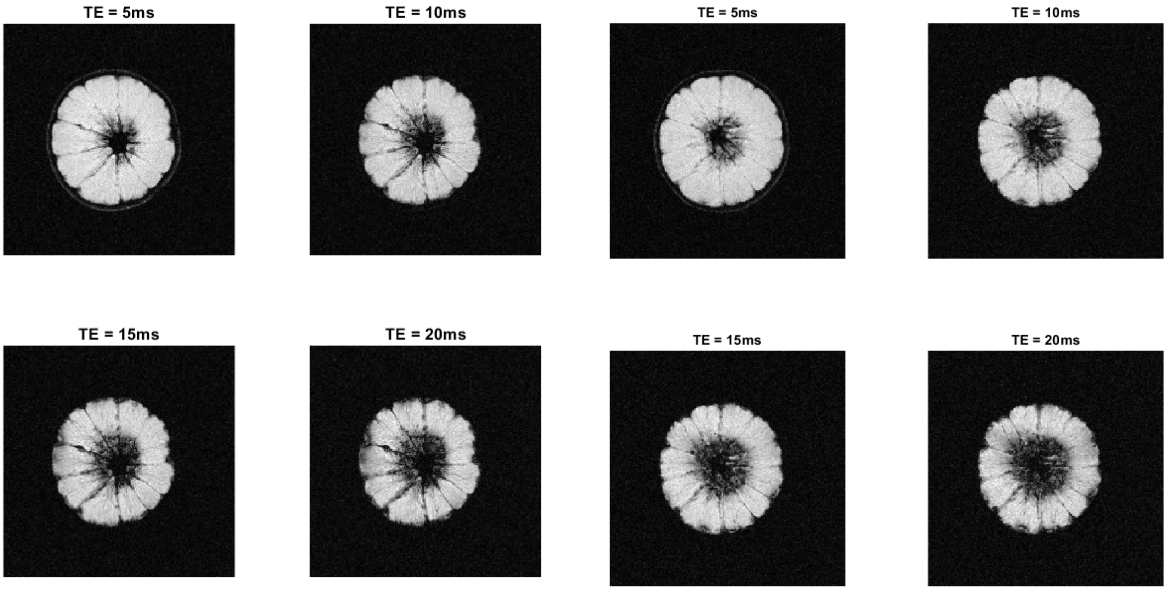
A Images for All Slices

Images for slices other than slice 3 are provided in this appendix.



(a) Slice 1 (b) Slice 2

Figure A1: Images with Different TE Values



(a) Slice 3 (b) Slice 4

Figure A2: Images with Different TE Values

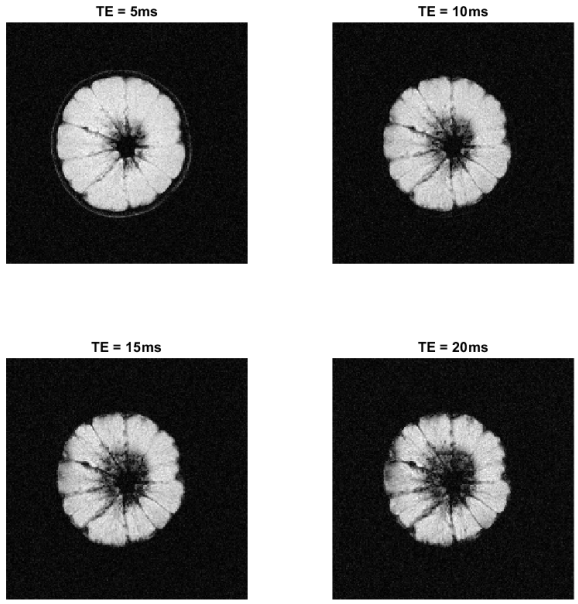
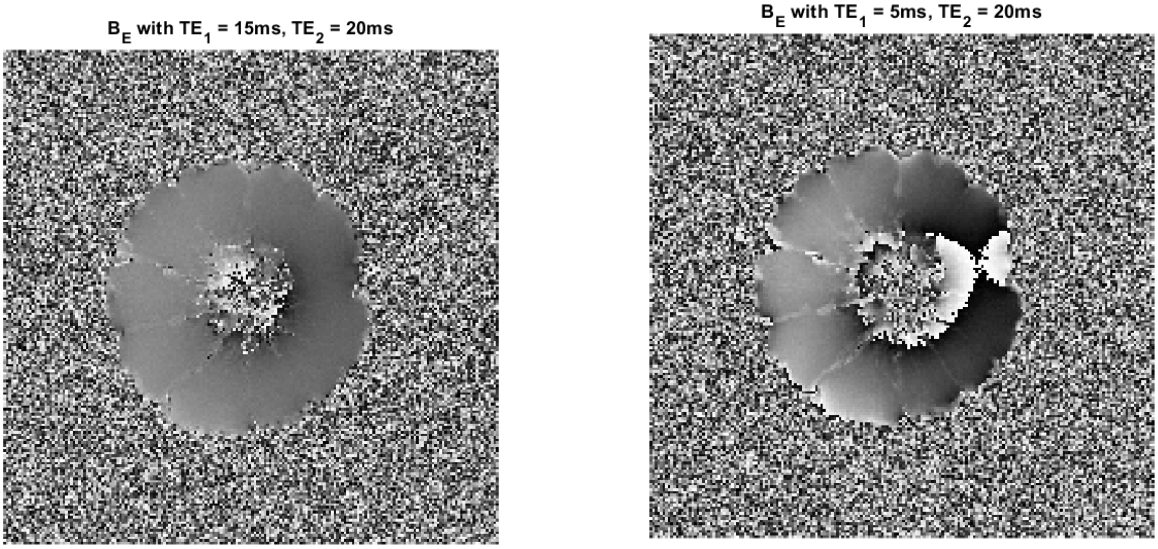


Figure A3: Images with Different TE Values, Slice: 5

B Additional B_0 Maps

Additional B_0 maps obtained using different echo-time pairs and results from gradient field mapping are included here.



(a) $TE_1 = 5$ ms, $TE_2 = 20$ ms

(b) $TE_1 = 15$ ms, $TE_2 = 20$ ms

Figure B1: B_0 Maps Obtained Using Different Echo-Time Pairs, Slice: 3

Gradient field mapping uses the same parameters except $TE_1 = 10$ ms, $TE_2 = 12.44$ ms.

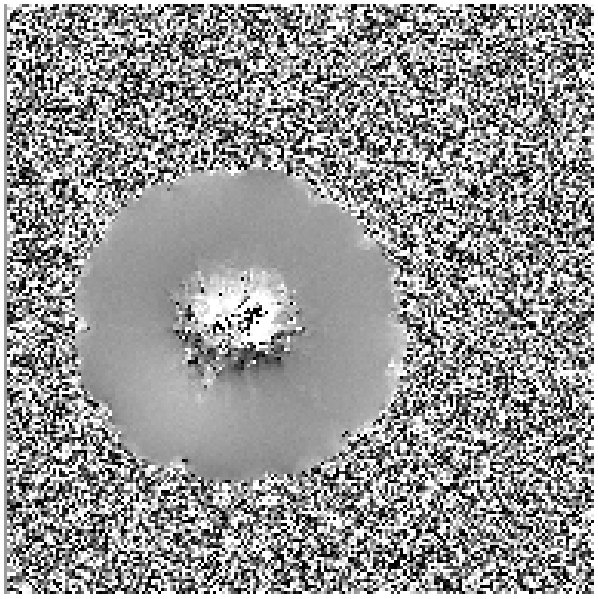


Figure B2: B_0 map from gradient field mapping, Slice: 3

C MATLAB Code for Processing

To read the .dat files, mapVBVD function [2] was used in MATLAB.

load data

```
slice = 5;
M1 = get_data('meas_MID50_gre_v1_FID34462.dat',slice);
M2 = get_data('meas_MID51_gre_v2_FID34463.dat',slice);
M3 = get_data('meas_MID52_gre_v3_FID34464.dat',slice);
M4 = get_data('meas_MID53_gre_v4_FID34465.dat',slice);

TE1=5*(10^(-3));
TE2=10*(10^(-3));
TE3=15*(10^(-3));
TE4=20*(10^(-3));
```

Arrayify

```
TEs = [TE1 TE2 TE3 TE4];
Ms =  {M1  M2  M3  M4 };
```

plot images:

```
for i = 1:4
    m_t = fftshift(ifft2(Ms{i}));
    subplot(2,2,i)
    imshow(abs(m_t),[])
    title(strcat("TE = ",num2str(TEs(i)*1000),"ms"))
end
set(gcf,'Color','white')
```

Find b0

```
clf;
i1 = 3;
i2 = 4;

gamma_bar = 42.58 * (10^6);
m1 = fftshift(ifft2(Ms{i1}));
m2 = fftshift(ifft2(Ms{i2}));
phase = angle(m1.*conj(m2));
fe = phase/(2*pi*(TEs(i1)-TEs(i2)));
b0 = fe/gamma_bar;
% imshow(b0,[])
imagesc(b0)
axis image
colormap parula
colorbar
title(strcat("B_E with TE_1 = ",num2str(TEs(i1)*1000),"ms, TE_2 =
",num2str(TEs(i2)*1000) ,"ms"))
set(gcf,'Color','white')
```

mask orange

```
clf;
```

```
imagesc(((abs(m1)>1*10^(-7)).*(b0+(14*10^-7)))-(14*10^-7))
axis image
colormap hot
colorbar
title(strcat("B_E with TE_1 = ",num2str(TEs(i1)*1000),"ms, TE_2 =
",num2str(TEs(i2)*1000) ,"ms"))
set(gcf,'Color','white')
```

noise analysis

```
clf;
mask = true(size(b0));           % start with everything
mask(35:150, 40:150) = false;    % exclude object region
noise_pixels = b0(mask);   % 1D vector automatically
var(noise_pixels)
subplot(1,2,1)
imshow(reshape([noise_pixels.' zeros([1 12])],[192 125]).',[])
title("Empty Voxel Noise ")
subplot(1,2,2)
histogram(noise_pixels, 20)
xlabel('B_0 (Tesla)')
ylabel('Count')
title('Background B_0 Noise Histogram')
set(gcf,'Color','white')
```

Noise analysis for actual image

```
clf;
actual_noise = m_t(mask)
var(actual_noise)
imshow(abs(reshape([actual_noise.' zeros([1 12])],[192 125])).',[])
```

Functions

```
function k2 = get_data(filename,slice)
    twix = mapVBVD(filename);
    % or mapVBVD(122) if file is in pwd :contentReference[oaicite:0]{index=0}
    % Helpful flags (optional)
    twix.image.flagRemoveOS = 1;           % remove 2x oversampling in read
direction :contentReference[oaicite:1]{index=1}
    twix.image.flagDoAverage = 0;          % keep averages unless you want them
averaged :contentReference[oaicite:2]{index=2}

    k = twix.image();                      % raw k-space
(complex) :contentReference[oaicite:3]{index=3}
    % k is typically: [Col, Cha, Lin, Par, Sli, Ave, Phs,
Eco, ...] :contentReference[oaicite:4]{index=4}

    % squeeze to [Col, Cha, Lin] for a 2D case (no partitions)
    k2 = squeeze(k(:,:,:,1, slice, 1, 1));   % adjust indices to your dataset
end
```